30/10/25

Chernoff bound

X : RV

$$\Pr(X > s) \leq \inf_{t>0} E[e^{tX}] \cdot e^{-st}$$

Ex: $X \sim N(0,1)$

$$\Pr(X > 10) \leq \overset{\text{Chernoff}}{e^{-50}} \quad \text{vs.} \quad \overset{\text{Markov}}{\text{not applicable}} \quad \text{vs.} \quad \overset{\text{Chebyshev}}{\frac{1}{100}}$$

$$\Pr(X < s) = \Pr(\enclose{circle}{-X} > -s) \leq \inf_{t>0} E[e^{-tX}] e^{st}$$

$$= \inf_{t<0} E[e^{tX}] e^{-st}$$

Sums of iid RVs

$Y = X_1 + \dots + X_n$ $X_i \sim X$ indep

$$\Pr(Y > s) \leq \inf_{t>0} E[e^{tY}] e^{-st}$$

$$= \inf_{t>0} E[e^{tX}]^n e^{-st}$$

Ex: $Y \sim \text{Binomial}(n, 1/2) = X_1 + \dots + X_n$

$X_i \sim \text{Bernoulli}(1/2)$ indep

$$\Pr\left(Y > \frac{n}{2} + \varepsilon\right) \leq \inf_{t>0} \left(\frac{1}{2} \cdot e^{t \cdot 0} + \frac{1}{2} e^{t \cdot 1}\right)^n e^{-\left(\frac{n}{2}+\varepsilon\right)t} = \left(\frac{1+e^t}{2}\right)^n e^{-\left(\frac{n}{2}+\varepsilon\right)t}$$

Set $t = 1/2$

$$\Pr\left(Y > \frac{n}{2} + \varepsilon\right) \leq \left(\frac{1+e^{1/2}}{2}\right)^n e^{-\left(\frac{n}{2}+\varepsilon\right)\frac{1}{2}}$$

$$= e^{n \log\left(\frac{1+e^{1/2}}{2}\right) - \frac{1}{2}\left(\frac{n}{2}+\varepsilon\right)}$$

$$= e^{-n\left[\frac{1}{4} + \frac{\varepsilon}{2n} - \underbrace{\log\left(\frac{1+e^{1/2}}{2}\right)}_{\approx 0.28}\right]}$$

$\varepsilon = n/4$

$$\Pr\left(Y > \frac{3n}{4}\right) \le e^{-n\left(\overbrace{\frac{1}{8} - 0.03}^{+ve}\right)}$$

Markov Chebyshev

vs. $\frac{2}{3}$ vs. order $1/n$

Chernoff for Binomial(n, p)

$Y = X_1 + \cdots + X_n$ $X_i \sim$ Bernoulli(p), indep.

$\mu_Y = np, \quad \sigma_Y^2 = np(1-p)$

$$\Pr(Y > np + \varepsilon) \le \inf_{t>0} \big(\underbrace{(1-p)e^{t\cdot 0} + pe^{t}}_{\downarrow}\big)^n e^{-(np+\varepsilon)t}$$

$1 + p(e^t - 1) < e^{p(e^t-1)}$ $1 + x \le e^x$ $(x > 0)$

$$= \inf_{t>0} e^{n\log(1+p(e^t-1)) - (np+\varepsilon)t}$$

$$\le \inf_{t>0} e^{np(e^t-1) - (np+\varepsilon)t}$$

$\varepsilon = \delta \cdot np$

$$= \inf_{t>0} e^{np(e^t - 1 - t(1+\delta))}$$

$e^{t^*} = 1 + \delta$

$t^* = \log(1+\delta)$

$$\Pr\big(Y > np(1+\delta)\big) \le e^{np(1+\delta-1-(1+\delta)\log(1+\delta))}$$

$$= e^{-np((1+\delta)\log(1+\delta) - \delta)}$$

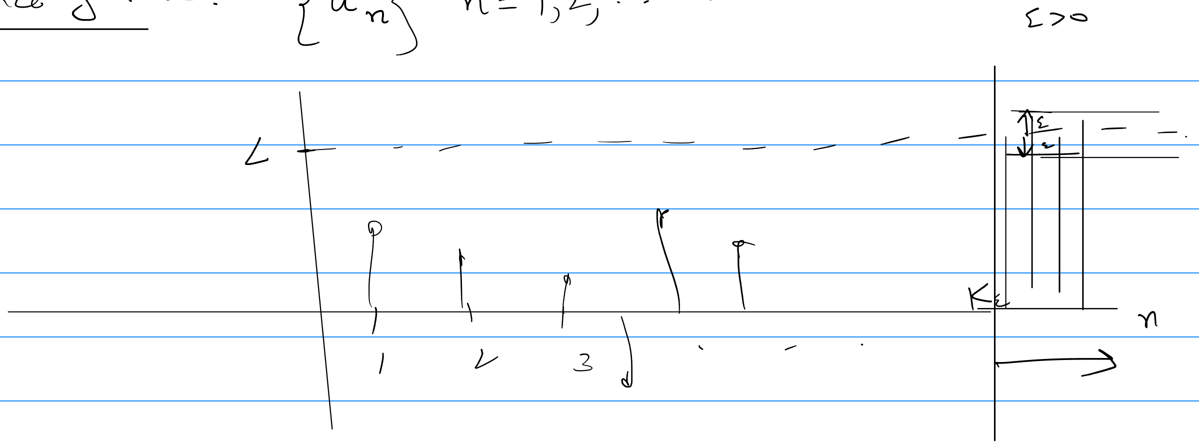
Convergence of RVs

$X_1, X_2, \ldots, X_n \sim$ iid X μ_X, σ_X^2

Sample mean: $\bar{X}_n = \dfrac{X_1 + \cdots + X_n}{n}$ RV

$E[\bar{X}_n] = \mu_X \qquad \mathrm{Var}(\bar{X}_n) = \dfrac{\sigma_X^2}{n}$

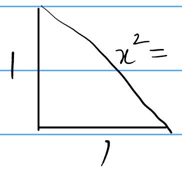
Sequence of reals: $\{a_n\}$ $n = 1, 2, \dots$



$$\lim_{n\to\infty} a_n = L$$

$\{a_n\}$

a_n : rational



$x^2 = 2$ not rational

$$\left(\lim_{n\to\infty} a_n\right)^2 = 2$$

Sequence of RVs (Some Probability Space)

$X_1, X_2, \dots, X_n, \dots$

↓

function : $\Omega \to \mathbb{R}$

→ Create a "meaningful" sequence of numbers whose convergence will imply something about convergence of X_n.

When does X_n converge to a constant b?

$\boxed{\text{Idea: } \Pr(X_n = b) = 1 \quad \forall\, n > K.}$ ✗ "not great"

$\varepsilon > 0$:

$$a_{n,\varepsilon} \triangleq \underbrace{\Pr\left(|X_n - b| > \varepsilon\right)}_{\text{real number}}$$

Suppose $\lim_{n\to\infty} a_{n,\varepsilon} = 0$ for $\varepsilon > 0$